BEAUTY & COSMETICS BRAND INTELLIGENCE PLATFORM

Complete Project Understanding & Interview Handbook

Dataset • Tools • Pipeline • NLP • Metrics • Data Modeling • Power BI • Interview Q&A

How to use this document
Read Sections 1–6 before an interview to understand the project. Memorize the 30-second and 2-minute explanations in Section 7. Use the technical Q&A and honesty checklist at the end to prepare for follow-up questions. Do not memorize claims you cannot explain.

This guide reflects the final Nykaa-review-based project as described in the project notes and dashboard summary. It is not a line-by-line reproduction of the repository. Verify exact implementation details in the current code before claiming them.

1. The project in one page

Project name: Beauty & Cosmetics Brand Intelligence Platform

The problem: Beauty brands receive large volumes of customer reviews. Reading them one by one makes it difficult to compare brands, understand the reasons behind ratings, spot recurring complaints, or turn feedback into decisions.

The solution: A Python analytics pipeline transforms review-level data into sentiment labels, customer-feedback themes, competitor metrics and structured BI outputs. Power BI presents those outputs in a dashboard designed for business users.

Project fact	What to remember
Dataset	Nykaa cosmetics and beauty product reviews; commonly associated with the Kaggle dataset "Cosmetics and Beauty Products Reviews – Top Brands."
Final processed volume	61,275 reviews
Brands	11
Date coverage	20 May 2013 – 22 October 2022
Mean rating	4.41
Core tools	Python, Pandas, NumPy, SQLite, scikit-learn, regex, Plotly, Power BI
Main outputs	Sentiment results, Voice of Customer topic reports, competitor scorecards, KPI exports, star-schema CSVs, Power BI dashboard
Dashboard pages	Executive Overview; Competitive Benchmarking; Voice of Customer Intelligence

The core story
I turned unstructured product reviews into a repeatable analytics pipeline and a Power BI decision-support dashboard. The value is not just showing ratings; it is connecting brand performance to the praise and complaints that may explain it.

2. Where the dataset came from

The final project uses a real Nykaa product-review dataset covering cosmetics and beauty brands. The dataset is commonly listed on Kaggle under the title "Cosmetics and Beauty Products Reviews – Top Brands," associated with Jithinanie Varghese.

Dataset link to check: https://www.kaggle.com/datasets/jithinanievarghese/cosmetics-and-beauty-products-reviews-top-brands

The raw data includes fields corresponding to review ID, brand, review title, review text, rating, review date and author. The project's ingestion layer maps these source-specific column names into standardized internal fields.

Important wording
Say: "The dataset contains reviews from Nykaa and was sourced from Kaggle." Do not imply that you collected the 61,275 reviews directly from Nykaa's live systems or that the dashboard shows the current market. The reviews span 2013–2022.

The initial dataset profile had 61,284 rows; the final integrated pipeline processed 61,275 records. The difference means the raw row count and final processed count are not the same. If asked about the exact exclusion rule for each of the nine rows, check the preprocessing code rather than guessing.

3. What tools we used and why

Tool / technology	Role in this project	How to explain it
Python	Pipeline orchestration and analytics	The main language connecting ingestion, text analysis, metrics and exports.
Pandas	Tabular data processing	Used to load CSVs, standardize columns, filter records, group data and export results.
NumPy	Numerical operations	Supports calculations and numerical handling in analytics.
SQLite	Structured local storage	Provides a local database for standardized data and downstream reads.
Regular expressions (regex)	Phrase matching for topic mapping	Allows multiword phrases such as "hair fall" or "white cast" to be matched as phrases rather than loose individual words.
scikit-learn / NMF	Topic-modeling option and supporting ML	NMF can represent documents using combinations of latent topics when the corpus and dependencies support it.

BERTopic	Optional topic-modeling path	The architecture supports BERTopic when installed; do not claim it was the method used for every final result without checking the run configuration.
Hugging Face Transformers / RoBERTa	Supported sentiment-model path	A transformer-based model was available as an option, but the final 61K-record run used the lexical fallback.
Custom cosmetics lexicon	Final large-run sentiment fallback	A faster local rule/lexicon approach designed around cosmetics and skincare vocabulary.
Plotly	Interactive figures	Used for generated analytical visualizations and HTML figures.
Power BI	Business-facing dashboard	Used to present KPIs, brand comparisons, trends and customer feedback themes.
Automated tests	Reliability checks	Tests cover areas such as ingestion, preprocessing, database, sentiment, services and BI metrics; a prior walkthrough reported 45 passing tests.

Tools that are not confirmed as part of the final dashboard pipeline should not be added to your interview list just because they are common in data projects. The list above focuses on the documented final implementation.

4. End-to-end architecture: what happens to a review?

Stage	Input	Work performed	Output
1. Ingestion	Raw Nykaa CSV	Read the file and map source column names to standard fields.	Standardized review records
2. Validation / preprocessing	Standardized records	Check required fields, handle missing values and prepare records for analysis.	Usable review dataset
3. Sentiment	Review text	Assign sentiment labels and downstream scores/confidence fields through the configured analyzer.	Record-level sentiment results
4. Voice of Customer	High-confidence positive/negative review text	Map text to meaningful praise and complaint categories; audit unmapped cases.	Topic assignments and topic reports
5. Business metrics	Reviews + sentiment + topics	Aggregate by brand, competitor, date and theme.	KPIs, scorecards and comparison tables
6. BI data modeling	Analytical tables	Export one review-level fact table and dimension tables.	Star-schema CSV exports
7. Dashboard	Structured exports	Build report pages and visuals in Power BI.	Interactive decision-support dashboard

4.1 Ingestion and standardization

The ingestion layer is designed to make data sources consistent. A source adapter handles the original schema and maps it into standard internal fields. For the Nykaa data, the mapping included:

Source field	Standardized field
review_id	source_record_id
brand_name	brand
review_title	title
review_text	text
review_rating	rating
review_date	created_at
author	author

Why this matters: downstream code can work against a consistent schema instead of depending on the exact column names of one CSV. The CSV source adapter was also updated to recognize review_rating as a rating field.

4.2 Storage and intermediate outputs

SQLite provides local structured storage. CSVs are used as inspectable intermediate and final artifacts, including record-level sentiment results, topic reports, scorecards and Power BI exports. This separation makes it easier to debug the pipeline and inspect what each stage produced.

4.3 Sentiment analysis

The sentiment component assigns a sentiment label to each review. The architecture supports a transformer-based route using CardiffNLP's twitter-roberta-base-sentiment-latest through Hugging Face Transformers. For the final large dataset

run, the configured path explicitly used force_fallback=True, so the final results came from the local cosmetics/skincare lexical analyzer.

- The lexical path is faster and avoids the compute overhead of running a transformer over every record.
- Lexical matching can be domain-aware, but it can miss context, sarcasm, negation and unusual wording.
- Do not call the final results "RoBERTa predictions." Say the project supports RoBERTa, but the 61,275-record run used the lexical fallback.

The sentiment output is stored at review level in data/sentiment_results.csv. The downstream model includes fields such as sentiment_score and sentiment_confidence; check their implementation before describing the exact scale or calibration.

4.4 Voice of Customer (VoC) and topic taxonomy

The VoC layer converts large numbers of individual reviews into recurring themes. The project supports an adaptive topic extraction chain: BERTopic when available, NMF when the corpus/dependencies support it, and a lexical taxonomy fallback in smaller or constrained cases. The final business taxonomy maps review language into interpretable categories.

Positive / praise themes	Negative / complaint themes
Visible Results	Hair Fall
Product Quality	Skin Irritation
Value For Money	Breakouts
Skin Hydration	Product Ineffectiveness
Effective Ingredients	Price Concerns
Good Packaging	Packaging Issues
Pleasant Fragrance	White Cast
Lightweight Formula	Strong Fragrance
Repurchase Intent	Delivery Issues

A key refinement was improving phrase matching. Instead of relying only on individual tokens, the mapping used regex word boundaries and phrase alternatives so expressions like "hair fall," "white cast," and "value for money" could be recognized. More specific categories were prioritized over generic categories such as Product Quality and Product Ineffectiveness.

What the 99.51% number means
After the taxonomy refactor, 233 of 47,991 high-confidence positive/negative records were unmapped. (47,991 − 233) / 47,991 = 99.51% mapping coverage. This is coverage of the taxonomy mapping step, not sentiment accuracy, topic-model accuracy, or coverage of all 61,275 reviews.

5. Business metrics: what they mean

The metric layer makes the project useful to a business audience. These definitions describe the documented project logic; project-specific composite scores may use custom weights or thresholds.

Metric	Definition / interpretation	Business question
Average Rating	Mean of available numeric ratings.	How highly do customers rate the products?
Positive Rate	Positive reviews ÷ total reviews in the chosen scope × 100.	What proportion of reviews are positive?
Negative Rate	Negative reviews ÷ total reviews in the chosen scope × 100.	What proportion are negative?
Net Sentiment	(Positive count − Negative count) ÷ total reviews × 100.	Is the overall balance of feedback positive or negative?
Visibility / review volume	Number of review records associated with a brand.	Which brand gets the most review activity in this dataset?
Share of Voice (SOV)	Brand review count ÷ all included brand review counts × 100.	What share of review volume does the brand have?
Sentiment gap	Net sentiment for comparator A minus net sentiment for comparator B.	Which of two selected brands has the stronger sentiment result?
Complaint / praise gap	Difference between comparator-level complaint or praise measures, according to the selected definition.	Where does one brand have more or fewer feedback issues/themes?
Topic diversity	Shannon entropy: $H = -\Sigma\, p_i \log_2(p_i)$, where p_i is a topic's share.	Is feedback spread across many topics or concentrated in a few?
Complaint concentration	Top complaint count ÷ total complaints × 100.	How much of the complaint burden comes from the biggest issue?

Data completeness	Non-null required-field cells ÷ expected required-field cells × 100.	How complete is the data for analysis?
Pipeline health	Successful recorded runs ÷ total recorded runs × 100.	How often does the pipeline run successfully, if run history is recorded?

Review-volume SOV is not sales market share. The dataset does not establish product sales, revenue, unique customers, or total category sales. Be explicit that the competitive analysis compares brands within the available review dataset.

The project also defines composite indices such as Brand Health Score, CSAT Index, Competitive Position Score and Data Quality. Treat these as project-defined metrics; verify the actual weights, thresholds and formula in the KPI definitions code before quoting them.

6. Data model and Power BI

6.1 Star schema

The BI export separates review-level facts from descriptive dimensions. The intended grain of fact_customer_feedback.csv is one row per review.

Table	Purpose	Examples of fields
fact_customer_feedback	Central review-level fact table	Review keys, rating, sentiment fields, text length, brand_key, topic_key, source_key, date_key
dim_brand	Brand attributes	Brand key and brand name
dim_topic	Topic attributes	Topic key and mapped/discovered topic labels
dim_source	Source attributes	Source key and source name
dim_date	Calendar attributes	Date key, year, month, month name, quarter, day of week

In a star schema, dimension tables connect one-to-many to the fact table. This model lets Power BI filter and aggregate review facts by brand, topic, source and time. The project notes report zero orphan records in the export audit, but re-run the validation if you need to make that claim as a current test result.

6.2 The three dashboard pages

Page	Main question	What it brings together
Executive Overview	What is happening?	Headline KPIs, overall rating/sentiment, review volume and trends, plus key findings.
Competitive Benchmarking	Who is winning?	Brand comparison, review-volume SOV, ratings, sentiment and comparative strengths/gaps.
Voice of Customer Intelligence	Why are they winning or losing?	Recurring praise and complaints, topic distributions, topic/sentiment relationships and customer pain points.

The final dashboard's purpose is to move from the overall picture, to brand comparison, to the feedback themes that help explain those comparisons.

7. Your interview explanation

7.1 30-second version

Say this
I built a Beauty & Cosmetics Brand Intelligence Platform using 61,275 Nykaa product reviews across 11 brands. I used Python to standardize the data, analyze sentiment, map customer feedback into cosmetics-specific praise and complaint themes, and calculate brand and competitor KPIs. I then exported the analytical data into a star schema and built a three-page Power BI dashboard covering executive performance, competitive benchmarking and Voice of Customer insights.

7.2 Two-minute version

The project addresses a practical problem: a large volume of product reviews contains useful customer feedback, but it is difficult to compare brands or understand recurring issues by reading reviews manually.

I used a Nykaa cosmetics review dataset sourced from Kaggle. The final pipeline processed 61,275 reviews across 11 brands, with dates spanning 2013 to 2022. I built a Python ingestion layer that standardized the original columns and prepared the data for analysis.

For sentiment, the architecture supports a transformer-based RoBERTa model and a local lexical fallback. For the final large dataset run, I used the fallback to process the data efficiently. I then built a Voice of Customer layer to map feedback into nine positive and nine negative categories. A taxonomy audit and phrase-matching refactor improved the mapping coverage to 99.51% across the 47,991 high-confidence positive and negative records; that figure is mapping coverage, not model accuracy.

Next, I calculated metrics such as net sentiment, positive and negative rates, review-volume Share of Voice, competitor gaps and topic concentration. I exported a review-level fact table with brand, topic, source and date dimensions, then used Power BI to create three pages: Executive Overview, Competitive Benchmarking and Voice of Customer Intelligence. The goal was to help a business user see what is happening, compare brands, and understand the customer themes behind the numbers.

7.3 If asked “What did you personally build?”

Answer with the components you can personally explain and demonstrate: the data ingestion/standardization pipeline, sentiment and VoC processing, taxonomy refinement, metric/scorecard generation, star-schema exports and Power BI report. Be precise about what you implemented yourself versus what the codebase supports as an optional path. If a component was generated or adapted with assistance, describe your actual role accurately.

7.4 If asked “What is the business value?”

- Reduces the effort required to summarize large volumes of review text.
- Allows consistent comparison of brands using the same review dataset and metric definitions.
- Connects numerical outcomes, such as rating or sentiment, to recurring praise and complaint themes.
- Helps identify potential areas for product, packaging, pricing, formula or customer-experience investigation.

Use “helps identify signals” or “supports decisions,” rather than claiming the analysis proves why a brand succeeds or establishes causal effects.

8. Findings from the final project

The final dashboard summary reported the following findings. These describe this dataset and its historical review window; they are not claims about current market share.

Finding	How to interpret it
Product Quality generated the most customer conversation.	Product-related experience was a dominant theme in the analyzed review text.
Lightweight Formula and Visible Results were associated with positive feedback.	These are examples of themes customers praised.
Product Ineffectiveness and Hair Fall were major dissatisfaction themes.	These complaint themes may be useful for deeper product/customer-experience investigation.
Nykaa Cosmetics led review volume / SOV.	It had the largest share of review records among the included brands; this is not proof of the largest sales share.
Kay Beauty had the highest average rating in the final dashboard summary.	Average rating and review volume are different dimensions; a brand can lead one without leading the other.

Historical notes also recorded Nykaa Cosmetics at approximately 28.80% review-volume SOV and Kay Beauty at approximately 22.49% in a dynamic top-two comparison. An earlier dashboard snapshot recorded Kay Beauty’s average rating at 4.70. Recalculate the metrics from the current exported files before quoting these exact numbers in an interview.

9. Challenges and what you learned

Challenge	What you did / what it taught you
Different source schemas	Used a standardized ingestion interface so downstream analysis did not depend on raw column names.
Processing a large review corpus	Used a faster local lexical fallback for the final large run instead of relying on transformer inference for every record.
Generic or poorly mapped topics	Audited unmapped samples and refined phrase matching and category priority.
Comparing brands fairly	Used consistent metric definitions and dynamically selected comparison brands based on available review volume instead of hard-coding absent brands.
Turning analysis into BI	Exported a fact table and dimensions so dashboard visuals could be filtered and aggregated consistently.
Distinguishing coverage from accuracy	Learned that a high taxonomy-mapping coverage rate does not prove the sentiment or topic labels are correct.

10. Likely recruiter questions and strong answers

Why did you choose this project?

I wanted to work on a problem where data analysis connects directly to business decisions. Beauty reviews contain opinions about product performance, price, packaging and customer experience. The project lets me connect NLP-style text processing with competitor analytics and a business-facing dashboard.

Why use Python and Power BI together?

Python handles repeatable data preparation, text processing and metric generation. Power BI is better suited to interactive exploration by business users. Separating the analytical pipeline from the presentation layer makes the workflow easier to rerun and inspect.

What was the dataset source?

The final project uses a Kaggle dataset of Nykaa cosmetics and beauty product reviews, commonly listed as Cosmetics and Beauty Products Reviews – Top Brands. The final integrated pipeline processed 61,275 records across 11 brands.

Did you scrape Nykaa yourself?

The final project described here is based on a downloaded review dataset. I should not describe it as a live scraper or direct Nykaa API integration unless I can demonstrate that in the final implementation.

Did you use RoBERTa?

The architecture supports a Hugging Face RoBERTa model, but the final 61,275-review run used the local lexical fallback. I distinguish the supported model path from the method actually used for the final run.

How does the sentiment fallback work?

It uses cosmetics/skincare-oriented lexical signals to assign sentiment labels quickly across a large number of reviews. It is efficient, but it is not equivalent to a context-aware transformer and should be evaluated against human-labelled examples before treating the output as ground truth.

How did you identify customer topics?

The system supports topic extraction paths such as BERTopic and NMF, plus a lexical taxonomy fallback. The final business layer maps feedback to interpretable categories like Visible Results, Lightweight Formula, Hair Fall and Product Ineffectiveness. I refined phrase matching to better handle multiword expressions.

Is 99.51% your model accuracy?

No. It is mapping coverage among 47,991 high-confidence positive and negative records. After the refactor, 233 remained unmapped. It does not measure whether the sentiment or topic label was correct.

What is Share of Voice here?

It is a brand's share of review volume within the dataset: that brand's review count divided by the total included review count. It is a proxy for review presence, not actual sales market share.

Why a star schema?

It separates review-level facts from descriptive dimensions like brand, topic, source and date. This structure makes relationships and filtering clearer in Power BI and supports consistent aggregations.

What was the hardest part?

A meaningful challenge was making the topic mapping useful rather than just producing labels. Generic categories can absorb too many reviews, while phrase-only matching can miss variants. Auditing unmapped samples and refining phrase matching improved taxonomy coverage and interpretability.

How would you improve it next?

I would create a manually labelled evaluation set and measure sentiment precision, recall and F1, plus topic-level precision/recall. I would also inspect false positives, document the taxonomy rules, add repeatable data-quality checks and refresh the dataset before making present-day market claims.

Can this tell us why a brand is successful?

It can show associations between brand performance and the topics appearing in customer feedback. It cannot prove causation. Product availability, review selection bias, marketing, price and other factors could also influence the results.

How do you know the analysis is reliable?

The project has tests across ingestion, preprocessing, database, sentiment, services and BI metrics, and a prior walkthrough reported 45 passing tests. For model reliability specifically, automated software tests are not enough; I would still need a human-labelled evaluation set and error analysis.

What are the project limitations?

The reviews are historical and may not represent the current market. Lexical sentiment can miss context. Topic mapping coverage is not label accuracy. Reviews may not represent all customers, and review volume is not sales share. Composite scores also depend on project-specific definitions and weights.

11. Technical terms in plain English

Term	Plain-English meaning
Pipeline	A sequence of steps that transforms raw data into useful outputs.
Ingestion	Reading data from its source into the project.
Preprocessing	Standardizing and preparing data before analysis.
Sentiment analysis	Estimating whether text expresses positive, negative or another sentiment.
Lexicon	A collection of words or phrases associated with particular signals, such as positive or negative sentiment.
Topic taxonomy	A defined set of business categories used to organize feedback.
Topic modeling	A method for discovering groups of words/documents that tend to occur together.
Regex / regular expression	A pattern-matching method for finding text expressions, including multiword phrases.
Fallback	An alternative implementation used when the preferred method is unavailable, too costly or unsuitable.
Fact table	The central table containing events/records to analyze; here, the intended grain is one row per review.
Dimension table	A table describing a fact, such as which brand, topic, source or date it belongs to.
Star schema	A data model with a central fact table linked to descriptive dimension tables.
KPI	A key performance indicator used to track a business outcome or operational measure.
Coverage	The proportion of records handled by a mapping or processing step; it is not necessarily correctness.
Precision / recall	Evaluation metrics that measure how often predictions are correct and how many relevant cases are found.

12. What not to overclaim

- Do not say the final 61,275 reviews were classified by RoBERTa; the final large run used the lexical fallback.
- Do not call 99.51% taxonomy mapping coverage "99.51% accuracy."
- Do not call review-volume Share of Voice "market share" or sales share.
- Do not claim the dataset reflects the current market; the review dates run through October 2022.
- Do not claim the project scraped live Nykaa data unless the final code demonstrates that.
- Do not invent exact weights for Brand Health Score, CSAT Index or Competitive Position Score. Check the KPI definitions.
- Do not state that every review received a topic assignment; the reported high-confidence positive/negative subset had 233 unmapped records.
- Do not claim a causal explanation for a brand's success based only on review patterns.
- Do not present the historical 45-test result as a fresh run unless the tests have been rerun.

13. Five-minute pre-interview revision sheet

Memorize these facts
61,275 final processed reviews • 11 brands • Nykaa reviews sourced from Kaggle • 2013–2022 • Python + Pandas + SQLite + NLP/regex + Plotly + Power BI • sentiment + Voice of Customer taxonomy + competitor KPIs + star schema • three Power BI pages.

Explain the project in this order

1. Business problem: too many reviews to interpret manually.
2. Data: Kaggle dataset of Nykaa beauty/cosmetics reviews.
3. Pipeline: ingest and standardize → sentiment → customer themes → KPIs → star schema → Power BI.
4. Business outputs: compare brand performance and understand praise/complaint themes.
5. Key caveat: historical review data, review-volume SOV is not sales share, and 99.51% is mapping coverage rather than accuracy.

If you remember only three things

- What you built: a repeatable data-to-insight pipeline, not merely a dashboard.
- Why it matters: it connects brand-level metrics to customer feedback themes.
- Where to be precise: sentiment fallback, mapping coverage, and the meaning of Share of Voice.

14. Source and verification notes

Dataset listing to verify: https://www.kaggle.com/datasets/jithinanievarghese/cosmetics-and-beauty-products-reviews-top-brands

LinkedIn project post supplied by the user: https://lnkd.in/p/giTSawDN

The project details in this handbook are based on the project notes and prior summary of the final dashboard. The exact current repository source code was not attached for a line-by-line audit. Before an interview that is likely to involve deep code questions, inspect the current implementations of the sentiment analyzer, topic extractor, KPI definitions, pipeline entry points and star-schema exporter. This will let you explain exact function names, arguments, formula weights and error handling without guessing.

Suggested source files to review in the repository:

- pipeline_real_data.py — how the Nykaa dataset enters the pipeline
- src/data_sources/csv_source.py — column mapping and standardization
- src/sentiment/sentiment_analyzer.py — actual sentiment methods and fallback logic

- src/topic_modeling/topic_extractor.py — topic extraction and taxonomy mapping
- src/metrics/kpi_definitions.py — exact KPI formulas, targets and thresholds
- src/metrics/star_schema.py — fact/dimension schemas and key construction
- pipeline_bi.py — how BI outputs are generated